

Phase I/II Operating Characteristics

Random model - Toxicity prior comparison - allocation = one_stage - IPDE alpha = 0 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 1						
DLT rate	0.3	0.35	0.4	0.45	0.5	
Efficacy rate	0.05	0.06	0.07	0.08	0.09	
Utility 2	-0.2	-0.3	-0.3	-0.4	-0.4	
Utility 3	31	29.6	28.2	26.8	25.4	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	37.6	16.3	3	1	0.7	41.4
U3 - Toxicity prior Beta(0.15, 0.85)	39.6	13.9	3	0.2	0.1	43.2
U2 - Toxicity prior Beta(0.3, 0.7)	37.4	19.6	2.9	0.6	0.8	38.7
U3 - Toxicity prior Beta(0.3, 0.7)	40.1	15.4	3.2	0.2	0.1	41
U2 - Toxicity prior Beta(0.5, 0.5)	37.1	19.1	4.1	1.1	0.6	38
U3 - Toxicity prior Beta(0.5, 0.5)	40.4	16.7	3.2	0.1	0.2	39.4
U2 - Toxicity prior Beta(1, 1)	36	18.9	4.2	1.8	0.7	38.4
U3 - Toxicity prior Beta(1, 1)	41.3	15.6	3.6	0.2	0.1	39.2
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	14	6.5	2.9	1.7	0.6	
U3 - Toxicity prior Beta(0.15, 0.85)	14.4	6.2	2.7	1.7	0.6	
U2 - Toxicity prior Beta(0.3, 0.7)	13.5	6.6	3.1	1.9	0.7	
U3 - Toxicity prior Beta(0.3, 0.7)	14	6.2	3	1.9	0.7	
U2 - Toxicity prior Beta(0.5, 0.5)	13	6.8	3.5	2	0.8	
U3 - Toxicity prior Beta(0.5, 0.5)	13.6	6.3	3.3	2	0.8	
U2 - Toxicity prior Beta(1, 1)	12.9	6.7	3.4	2	0.9	
U3 - Toxicity prior Beta(1, 1)	13.7	6.2	3.1	2	0.8	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	1.2	3.4	0.8	0.7	0.2	
U3 - Toxicity prior Beta(0.15, 0.85)	1.2	3.2	0.8	0.7	0.2	
U2 - Toxicity prior Beta(0.3, 0.7)	1.2	3.2	0.9	0.7	0.2	
U3 - Toxicity prior Beta(0.3, 0.7)	1.3	3	0.9	0.7	0.2	
U2 - Toxicity prior Beta(0.5, 0.5)	1.2	3	0.8	0.8	0.2	
U3 - Toxicity prior Beta(0.5, 0.5)	1.2	2.8	0.8	0.7	0.2	
U2 - Toxicity prior Beta(1, 1)	1.2	2.5	0.8	0.6	0.1	
U3 - Toxicity prior Beta(1, 1)	1.2	2.4	0.8	0.6	0.1	
Scenario 2						
DLT rate	0.15	0.2	0.3	0.35	0.45	
Efficacy rate	0.2	0.25	0.5	0.5	0.5	
Utility 2	0.1	0.1	0.2	0.1	0.1	
Utility 3	46	47	58	56	52	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	25.6	17.2	38.5	14.1	3.7	0.9
U3 - Toxicity prior Beta(0.15, 0.85)	31.8	17.4	34.9	12.3	2.6	1
U2 - Toxicity prior Beta(0.3, 0.7)	24.3	16.7	38.4	15.8	3.8	1
U3 - Toxicity prior Beta(0.3, 0.7)	29.8	17.9	36.6	12.2	2.7	0.8
U2 - Toxicity prior Beta(0.5, 0.5)	23.9	15.9	39.6	15.5	4.5	0.6
U3 - Toxicity prior Beta(0.5, 0.5)	28.1	16.6	39.1	12.9	2.6	0.7
U2 - Toxicity prior Beta(1, 1)	23.4	16	38.8	17.1	4	0.7
U3 - Toxicity prior Beta(1, 1)	28.1	16.8	36.7	14.7	2.9	0.8
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	8.9	6.8	8.8	4.1	1.3	
U3 - Toxicity prior Beta(0.15, 0.85)	9.4	6.9	8.7	3.8	1	
U2 - Toxicity prior Beta(0.3, 0.7)	8.4	6.7	9.1	4.3	1.4	
U3 - Toxicity prior Beta(0.3, 0.7)	9	6.8	8.9	4.1	1.1	
U2 - Toxicity prior Beta(0.5, 0.5)	7.9	6.6	9.2	4.6	1.6	
U3 - Toxicity prior Beta(0.5, 0.5)	8.5	6.7	9.3	4.2	1.2	
U2 - Toxicity prior Beta(1, 1)	8	6.4	9	4.8	1.6	
U3 - Toxicity prior Beta(1, 1)	8.5	6.6	9.1	4.4	1.2	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.5	3.1	1.7	1	0.3	
U3 - Toxicity prior Beta(0.15, 0.85)	0.6	3.1	1.6	0.9	0.2	
U2 - Toxicity prior Beta(0.3, 0.7)	0.5	3	1.6	1	0.3	
U3 - Toxicity prior Beta(0.3, 0.7)	0.6	3	1.5	0.9	0.2	
U2 - Toxicity prior Beta(0.5, 0.5)	0.5	2.9	1.5	1	0.3	
U3 - Toxicity prior Beta(0.5, 0.5)	0.5	2.8	1.5	0.9	0.2	
U2 - Toxicity prior Beta(1, 1)	0.5	2.6	1.5	0.9	0.3	
U3 - Toxicity prior Beta(1, 1)	0.6	2.6	1.5	0.8	0.2	

Phase I/II Operating Characteristics

Random model - Toxicity prior comparison - allocation = one_stage - IPDE alpha = 0 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 3						
DLT rate	0.15	0.2	0.3	0.35	0.45	
Efficacy rate	0.1	0.4	0.4	0.4	0.4	
Utility 2	-0.1	0.2	0.1	0.1	-0.1	
Utility 3	40	56	52	50	46	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	11.2	44.7	26.8	12.1	3	2.2
U3 - Toxicity prior Beta(0.15, 0.85)	13.7	50.2	22	9.9	1.8	2.4
U2 - Toxicity prior Beta(0.3, 0.7)	10.8	43.6	27.4	13.4	2.8	2
U3 - Toxicity prior Beta(0.3, 0.7)	13.2	50.6	22.1	10.3	1.7	2.1
U2 - Toxicity prior Beta(0.5, 0.5)	9.8	43.7	28.4	12.9	3.3	1.9
U3 - Toxicity prior Beta(0.5, 0.5)	12.3	50.5	22.7	10.8	1.9	1.8
U2 - Toxicity prior Beta(1, 1)	9.3	44.9	27.5	12.9	3.7	1.7
U3 - Toxicity prior Beta(1, 1)	12	50.7	21.9	11.3	2.3	1.8
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	6.3	10.6	7.2	4.2	1.5	
U3 - Toxicity prior Beta(0.15, 0.85)	6.5	11.5	6.7	3.8	1.2	
U2 - Toxicity prior Beta(0.3, 0.7)	6.1	10.4	7.2	4.5	1.6	
U3 - Toxicity prior Beta(0.3, 0.7)	6.3	11.3	6.7	4.1	1.3	
U2 - Toxicity prior Beta(0.5, 0.5)	5.7	10.4	7.3	4.6	1.8	
U3 - Toxicity prior Beta(0.5, 0.5)	5.9	11.3	6.9	4.3	1.5	
U2 - Toxicity prior Beta(1, 1)	5.8	10.1	7.3	4.7	1.9	
U3 - Toxicity prior Beta(1, 1)	5.9	11.2	6.7	4.3	1.6	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.4	3.6	1.8	1.2	0.3	
U3 - Toxicity prior Beta(0.15, 0.85)	0.4	3.6	1.6	1.1	0.3	
U2 - Toxicity prior Beta(0.3, 0.7)	0.4	3.5	1.7	1.2	0.3	
U3 - Toxicity prior Beta(0.3, 0.7)	0.5	3.5	1.6	1.1	0.3	
U2 - Toxicity prior Beta(0.5, 0.5)	0.4	3.3	1.6	1.2	0.4	
U3 - Toxicity prior Beta(0.5, 0.5)	0.4	3.3	1.6	1.1	0.3	
U2 - Toxicity prior Beta(1, 1)	0.4	3.1	1.6	1.1	0.3	
U3 - Toxicity prior Beta(1, 1)	0.4	3.1	1.4	1	0.3	
Scenario 4						
DLT rate	0.05	0.1	0.18	0.3	0.4	
Efficacy rate	0.05	0.1	0.25	0.5	0.55	
Utility 2	0	0	0.1	0.2	0.1	
Utility 3	41	42	47.8	58	57	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	7.2	6.1	15.9	50.9	19.7	0.2
U3 - Toxicity prior Beta(0.15, 0.85)	11.8	7.5	14.4	49.9	16.2	0.2
U2 - Toxicity prior Beta(0.3, 0.7)	7	4.6	16.2	50.8	21.2	0.2
U3 - Toxicity prior Beta(0.3, 0.7)	10.8	6.5	15.4	50.2	16.9	0.2
U2 - Toxicity prior Beta(0.5, 0.5)	6.2	4.6	15.8	51.3	21.9	0.2
U3 - Toxicity prior Beta(0.5, 0.5)	9.6	5.3	16.1	50.5	18.3	0.2
U2 - Toxicity prior Beta(1, 1)	6.6	4	15.6	50.4	23.2	0.2
U3 - Toxicity prior Beta(1, 1)	9	6.1	14.5	51.2	19	0.2
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	4.1	4.3	6.4	10.3	4.8	
U3 - Toxicity prior Beta(0.15, 0.85)	4.4	4.4	6.5	10.5	4.2	
U2 - Toxicity prior Beta(0.3, 0.7)	4	4.1	6.4	10.4	5.1	
U3 - Toxicity prior Beta(0.3, 0.7)	4.2	4.3	6.4	10.7	4.3	
U2 - Toxicity prior Beta(0.5, 0.5)	3.8	4.1	6.3	10.5	5.3	
U3 - Toxicity prior Beta(0.5, 0.5)	4.1	4.2	6.4	10.9	4.5	
U2 - Toxicity prior Beta(1, 1)	3.8	4.1	6	10.7	5.5	
U3 - Toxicity prior Beta(1, 1)	4	4.1	6.2	10.9	4.8	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.2	2.9	1.5	2.4	0.9	
U3 - Toxicity prior Beta(0.15, 0.85)	0.3	2.9	1.5	2.4	0.8	
U2 - Toxicity prior Beta(0.3, 0.7)	0.2	2.8	1.4	2.4	0.9	
U3 - Toxicity prior Beta(0.3, 0.7)	0.2	2.9	1.4	2.4	0.8	
U2 - Toxicity prior Beta(0.5, 0.5)	0.2	2.8	1.4	2.3	0.9	
U3 - Toxicity prior Beta(0.5, 0.5)	0.2	2.8	1.4	2.3	0.8	
U2 - Toxicity prior Beta(1, 1)	0.2	2.7	1.4	2.2	0.8	
U3 - Toxicity prior Beta(1, 1)	0.2	2.7	1.4	2.2	0.8	

Phase I/II Operating Characteristics

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	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 5						
DLT rate	0.07	0.12	0.17	0.22	0.3	
Efficacy rate	0.05	0.3	0.3	0.3	0.3	
Utility 2	0	0.2	0.1	0.1	0	
Utility 3	40.2	53.2	51.2	49.2	46	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	6.8	32.8	27.7	21.4	10.9	0.4
U3 - Toxicity prior Beta(0.15, 0.85)	10	39.3	25.3	18.1	6.9	0.4
U2 - Toxicity prior Beta(0.3, 0.7)	6.5	30.5	28.9	22.3	11.4	0.4
U3 - Toxicity prior Beta(0.3, 0.7)	9.6	39.8	24.1	18.9	7.2	0.4
U2 - Toxicity prior Beta(0.5, 0.5)	6.7	30.7	29.8	21.5	10.9	0.4
U3 - Toxicity prior Beta(0.5, 0.5)	9.4	38.3	25.5	18.6	7.9	0.3
U2 - Toxicity prior Beta(1, 1)	6	31.1	28.3	22.5	11.6	0.5
U3 - Toxicity prior Beta(1, 1)	9.9	37.8	25.1	18.7	8.1	0.4
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	3.8	6.6	7.1	7.4	5.1	
U3 - Toxicity prior Beta(0.15, 0.85)	3.9	7.5	7.1	7.1	4.4	
U2 - Toxicity prior Beta(0.3, 0.7)	3.7	6.4	7.2	7.4	5.2	
U3 - Toxicity prior Beta(0.3, 0.7)	3.8	7.3	7.1	7.2	4.5	
U2 - Toxicity prior Beta(0.5, 0.5)	3.6	6.3	7.1	7.6	5.4	
U3 - Toxicity prior Beta(0.5, 0.5)	3.7	7.3	7.2	7.2	4.6	
U2 - Toxicity prior Beta(1, 1)	3.6	6.3	7	7.5	5.5	
U3 - Toxicity prior Beta(1, 1)	3.7	7	7	7.4	4.8	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.2	3	1.9	2.4	1.1	
U3 - Toxicity prior Beta(0.15, 0.85)	0.2	3.1	1.9	2.2	1	
U2 - Toxicity prior Beta(0.3, 0.7)	0.1	3	1.9	2.4	1.1	
U3 - Toxicity prior Beta(0.3, 0.7)	0.2	3.1	1.8	2.2	1	
U2 - Toxicity prior Beta(0.5, 0.5)	0.1	2.9	1.8	2.4	1.1	
U3 - Toxicity prior Beta(0.5, 0.5)	0.1	3	1.9	2.2	1	
U2 - Toxicity prior Beta(1, 1)	0.1	2.8	1.9	2.3	1.1	
U3 - Toxicity prior Beta(1, 1)	0.2	2.9	1.9	2.3	1	
Scenario 6						
DLT rate	0.29	0.34	0.38	0.43	0.52	
Efficacy rate	0.3	0.32	0.35	0.38	0.4	
Utility 2	0	0	0	-0.1	-0.1	
Utility 3	46.4	45.6	45.8	45.6	43.2	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	61.7	18.3	5.5	2	0.5	12
U3 - Toxicity prior Beta(0.15, 0.85)	66.3	16.1	3.6	2	0.2	11.8
U2 - Toxicity prior Beta(0.3, 0.7)	60.7	19.2	5.8	2.7	0.6	11
U3 - Toxicity prior Beta(0.3, 0.7)	65.6	16.1	4.7	2	0.5	11.1
U2 - Toxicity prior Beta(0.5, 0.5)	58.8	19.6	7.1	2.7	0.7	11.1
U3 - Toxicity prior Beta(0.5, 0.5)	65.6	15.9	4.9	2.3	0.4	10.9
U2 - Toxicity prior Beta(1, 1)	58.1	20	7.2	2.5	1	11.2
U3 - Toxicity prior Beta(1, 1)	63.8	18.2	4.6	2.3	0.4	10.7
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	16.7	6.2	3	1.5	0.4	
U3 - Toxicity prior Beta(0.15, 0.85)	17.5	6	2.6	1.4	0.4	
U2 - Toxicity prior Beta(0.3, 0.7)	16.2	6.3	3.2	1.7	0.6	
U3 - Toxicity prior Beta(0.3, 0.7)	16.8	6.2	2.9	1.6	0.5	
U2 - Toxicity prior Beta(0.5, 0.5)	15.4	6.5	3.5	1.9	0.7	
U3 - Toxicity prior Beta(0.5, 0.5)	16.3	6.4	3.1	1.6	0.6	
U2 - Toxicity prior Beta(1, 1)	15.4	6.4	3.5	1.9	0.7	
U3 - Toxicity prior Beta(1, 1)	16.3	6.3	3.1	1.7	0.6	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.7	2.6	0.7	0.4	0.1	
U3 - Toxicity prior Beta(0.15, 0.85)	0.8	2.4	0.6	0.4	0.1	
U2 - Toxicity prior Beta(0.3, 0.7)	0.8	2.5	0.8	0.5	0.1	
U3 - Toxicity prior Beta(0.3, 0.7)	0.8	2.4	0.7	0.4	0.1	
U2 - Toxicity prior Beta(0.5, 0.5)	0.7	2.3	0.7	0.4	0.1	
U3 - Toxicity prior Beta(0.5, 0.5)	0.8	2.3	0.6	0.4	0.1	
U2 - Toxicity prior Beta(1, 1)	0.7	1.9	0.7	0.4	0.1	
U3 - Toxicity prior Beta(1, 1)	0.7	1.9	0.6	0.3	0.1	

Phase I/II Operating Characteristics

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	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 7						
DLT rate	0.16	0.2	0.26	0.31	0.36	
Efficacy rate	0.2	0.5	0.5	0.5	0.5	
Utility 2	0	0.3	0.2	0.2	0.1	
Utility 3	45.6	62	59.6	57.6	55.6	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	13.2	44.7	24.3	12.4	4.5	0.9
U3 - Toxicity prior Beta(0.15, 0.85)	16.6	48.7	21.8	9.6	2.4	0.9
U2 - Toxicity prior Beta(0.3, 0.7)	13.1	43.4	23.6	14.7	4.4	0.8
U3 - Toxicity prior Beta(0.3, 0.7)	16	48.4	21.3	10.5	2.9	0.9
U2 - Toxicity prior Beta(0.5, 0.5)	11.8	41.8	25.6	14.6	5.4	0.8
U3 - Toxicity prior Beta(0.5, 0.5)	14.5	48	23	10.5	3.2	0.8
U2 - Toxicity prior Beta(1, 1)	10.8	43.4	24.8	14.6	5.4	1
U3 - Toxicity prior Beta(1, 1)	13.2	49	22.7	11.6	2.8	0.7
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	7.4	11.7	6.1	3.4	1.3	
U3 - Toxicity prior Beta(0.15, 0.85)	7.7	12.5	5.9	2.9	0.9	
U2 - Toxicity prior Beta(0.3, 0.7)	7	11.6	6.4	3.6	1.3	
U3 - Toxicity prior Beta(0.3, 0.7)	7.4	12.5	5.9	3.1	1	
U2 - Toxicity prior Beta(0.5, 0.5)	6.6	11.6	6.5	3.6	1.5	
U3 - Toxicity prior Beta(0.5, 0.5)	6.9	12.5	6.1	3.2	1.1	
U2 - Toxicity prior Beta(1, 1)	6.6	11.5	6.5	3.7	1.5	
U3 - Toxicity prior Beta(1, 1)	6.9	12.5	6.1	3.3	1.1	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.3	3.3	1.3	0.8	0.3	
U3 - Toxicity prior Beta(0.15, 0.85)	0.3	3.4	1.2	0.6	0.2	
U2 - Toxicity prior Beta(0.3, 0.7)	0.3	3.2	1.3	0.8	0.3	
U3 - Toxicity prior Beta(0.3, 0.7)	0.3	3.3	1.1	0.7	0.2	
U2 - Toxicity prior Beta(0.5, 0.5)	0.3	3	1.2	0.8	0.3	
U3 - Toxicity prior Beta(0.5, 0.5)	0.3	3	1.1	0.6	0.2	
U2 - Toxicity prior Beta(1, 1)	0.3	2.8	1.1	0.8	0.2	
U3 - Toxicity prior Beta(1, 1)	0.3	2.8	1.1	0.6	0.2	
Scenario 8						
DLT rate	0.32	0.37	0.42	0.48	0.52	
Efficacy rate	0.1	0.15	0.2	0.3	0.4	
Utility 2	-0.2	-0.2	-0.2	-0.2	-0.1	
Utility 3	33.2	34.2	35.2	38.8	43.2	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	46.7	15.1	2.8	1.5	0.7	33.2
U3 - Toxicity prior Beta(0.15, 0.85)	47.2	14	3.1	1	0.8	33.9
U2 - Toxicity prior Beta(0.3, 0.7)	46.1	15.6	4.7	1.4	0.8	31.4
U3 - Toxicity prior Beta(0.3, 0.7)	47.3	15.1	3.8	0.9	0.7	32.2
U2 - Toxicity prior Beta(0.5, 0.5)	46.8	14.5	6.2	1.4	1.1	30
U3 - Toxicity prior Beta(0.5, 0.5)	47.6	15.4	5.1	0.8	0.8	30.3
U2 - Toxicity prior Beta(1, 1)	44.8	15.2	6.7	1.4	1.4	30.5
U3 - Toxicity prior Beta(1, 1)	47.5	15.8	4.6	1.2	0.8	30.1
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	15	6.2	2.7	1.4	0.5	
U3 - Toxicity prior Beta(0.15, 0.85)	15.5	6	2.5	1.4	0.4	
U2 - Toxicity prior Beta(0.3, 0.7)	14.5	6.4	3	1.7	0.6	
U3 - Toxicity prior Beta(0.3, 0.7)	14.9	6.2	2.8	1.6	0.5	
U2 - Toxicity prior Beta(0.5, 0.5)	13.9	6.5	3.4	1.8	0.7	
U3 - Toxicity prior Beta(0.5, 0.5)	14.3	6.4	3.2	1.8	0.6	
U2 - Toxicity prior Beta(1, 1)	14	6.3	3.3	1.9	0.7	
U3 - Toxicity prior Beta(1, 1)	14.6	6.2	3.1	1.8	0.7	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.9	2.9	0.7	0.5	0.1	
U3 - Toxicity prior Beta(0.15, 0.85)	1	2.8	0.7	0.5	0.1	
U2 - Toxicity prior Beta(0.3, 0.7)	1	2.8	0.7	0.5	0.1	
U3 - Toxicity prior Beta(0.3, 0.7)	1	2.7	0.7	0.5	0.1	
U2 - Toxicity prior Beta(0.5, 0.5)	0.9	2.5	0.7	0.5	0.1	
U3 - Toxicity prior Beta(0.5, 0.5)	0.9	2.5	0.7	0.5	0.1	
U2 - Toxicity prior Beta(1, 1)	0.9	2.1	0.7	0.4	0.1	
U3 - Toxicity prior Beta(1, 1)	0.9	2	0.7	0.4	0.1	

Phase I/II Operating Characteristics

Random model - Toxicity prior comparison - allocation = one_stage - IPDE alpha = 0 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 9						
DLT rate	0.15	0.18	0.22	0.26	0.29	
Efficacy rate	0.05	0.08	0.2	0.25	0.5	
Utility 2	-0.1	-0.1	0	0	0.2	
Utility 3	37	37.6	43.2	44.6	58.4	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	14.6	13.1	21.4	14.7	32.4	3.8
U3 - Toxicity prior Beta(0.15, 0.85)	22.5	15.1	18.9	13.1	26.5	3.9
U2 - Toxicity prior Beta(0.3, 0.7)	15	11.6	20.6	15.7	33.6	3.5
U3 - Toxicity prior Beta(0.3, 0.7)	22.2	13.9	18.4	14.3	27.7	3.5
U2 - Toxicity prior Beta(0.5, 0.5)	13.8	12	19.8	15.3	36.1	3
U3 - Toxicity prior Beta(0.5, 0.5)	21.8	14	17.8	13.8	29.2	3.4
U2 - Toxicity prior Beta(1, 1)	14.2	10.9	18.9	16.5	36.5	3
U3 - Toxicity prior Beta(1, 1)	19.8	14.4	17.4	13.5	31.4	3.5
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	6.6	5.7	5.9	5.4	6	
U3 - Toxicity prior Beta(0.15, 0.85)	7.2	5.6	5.8	5.4	5.6	
U2 - Toxicity prior Beta(0.3, 0.7)	6.2	5.5	5.9	5.6	6.4	
U3 - Toxicity prior Beta(0.3, 0.7)	6.9	5.5	5.9	5.6	5.9	
U2 - Toxicity prior Beta(0.5, 0.5)	5.8	5.4	5.9	5.7	6.8	
U3 - Toxicity prior Beta(0.5, 0.5)	6.4	5.3	5.9	5.8	6.2	
U2 - Toxicity prior Beta(1, 1)	5.8	5.3	5.9	5.7	7	
U3 - Toxicity prior Beta(1, 1)	6.4	5.2	5.7	5.7	6.6	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.6	3.4	1.6	1.9	1	
U3 - Toxicity prior Beta(0.15, 0.85)	0.6	3.3	1.6	1.9	0.9	
U2 - Toxicity prior Beta(0.3, 0.7)	0.5	3.3	1.5	2	1	
U3 - Toxicity prior Beta(0.3, 0.7)	0.6	3.3	1.5	1.9	0.9	
U2 - Toxicity prior Beta(0.5, 0.5)	0.5	3.1	1.5	1.9	1	
U3 - Toxicity prior Beta(0.5, 0.5)	0.5	3	1.5	1.9	0.9	
U2 - Toxicity prior Beta(1, 1)	0.5	2.8	1.5	1.9	0.9	
U3 - Toxicity prior Beta(1, 1)	0.6	2.8	1.6	1.8	0.8	
Scenario 10						
DLT rate	0.23	0.28	0.32	0.39	0.43	
Efficacy rate	0.05	0.4	0.42	0.45	0.5	
Utility 2	-0.2	0.1	0.1	0.1	0.1	
Utility 3	33.8	52.8	52.4	51.4	52.8	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	11.2	41.9	17.9	7.6	3	18.4
U3 - Toxicity prior Beta(0.15, 0.85)	12.9	44.7	16.1	6.4	1.5	18.4
U2 - Toxicity prior Beta(0.3, 0.7)	10	42.2	19.6	8	3.5	16.7
U3 - Toxicity prior Beta(0.3, 0.7)	12.4	44.7	16.9	6.5	2.2	17.3
U2 - Toxicity prior Beta(0.5, 0.5)	10.5	42.1	20.5	8.3	3.4	15.2
U3 - Toxicity prior Beta(0.5, 0.5)	11.6	45.8	17.3	6.9	2.5	15.9
U2 - Toxicity prior Beta(1, 1)	9.8	41.9	19.6	8.9	3.8	16
U3 - Toxicity prior Beta(1, 1)	11.2	45.7	17.5	6.6	2.4	16.6
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	8.9	10.3	5.3	2.7	1	
U3 - Toxicity prior Beta(0.15, 0.85)	9.1	11	5	2.4	0.8	
U2 - Toxicity prior Beta(0.3, 0.7)	8.6	10.2	5.5	2.8	1.1	
U3 - Toxicity prior Beta(0.3, 0.7)	8.8	10.7	5.2	2.7	0.9	
U2 - Toxicity prior Beta(0.5, 0.5)	7.9	10.4	5.9	3	1.2	
U3 - Toxicity prior Beta(0.5, 0.5)	8.1	10.8	5.7	2.8	1	
U2 - Toxicity prior Beta(1, 1)	8	10.3	5.6	3.1	1.4	
U3 - Toxicity prior Beta(1, 1)	8.2	10.8	5.4	2.8	1	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.5	3.9	1.2	0.8	0.2	
U3 - Toxicity prior Beta(0.15, 0.85)	0.6	3.8	1.1	0.7	0.2	
U2 - Toxicity prior Beta(0.3, 0.7)	0.5	3.7	1.1	0.8	0.2	
U3 - Toxicity prior Beta(0.3, 0.7)	0.6	3.7	1.1	0.7	0.2	
U2 - Toxicity prior Beta(0.5, 0.5)	0.5	3.3	1.2	0.8	0.2	
U3 - Toxicity prior Beta(0.5, 0.5)	0.6	3.3	1.1	0.7	0.1	
U2 - Toxicity prior Beta(1, 1)	0.5	2.8	1.1	0.7	0.2	
U3 - Toxicity prior Beta(1, 1)	0.5	2.9	1.1	0.6	0.1	

Phase I/II Operating Characteristics

Random model - Toxicity prior comparison - allocation = one_stage - IPDE alpha = 0 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 11						
DLT rate	0.22	0.32	0.44	0.59	0.67	
Efficacy rate	0.2	0.2	0.2	0.2	0.2	
Utility 2	0	-0.1	-0.2	-0.4	-0.5	
Utility 3	43.2	39.2	34.4	28.4	25.2	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	65.8	22.9	5.6	0.8	0	4.9
U3 - Toxicity prior Beta(0.15, 0.85)	70.3	21.1	3.3	0.6	0	4.7
U2 - Toxicity prior Beta(0.3, 0.7)	63.7	25.1	6	1	0	4.2
U3 - Toxicity prior Beta(0.3, 0.7)	67.8	22.1	4.8	0.8	0	4.5
U2 - Toxicity prior Beta(0.5, 0.5)	62.6	25.9	6.1	1.1	0.1	4.2
U3 - Toxicity prior Beta(0.5, 0.5)	67.9	22.6	4.3	0.7	0.1	4.4
U2 - Toxicity prior Beta(1, 1)	62.5	25.2	6.6	1.1	0.1	4.5
U3 - Toxicity prior Beta(1, 1)	66.8	23.2	4.2	1.1	0.1	4.6
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	15.7	7.7	3.8	1.7	0.4	
U3 - Toxicity prior Beta(0.15, 0.85)	16.4	7.6	3.5	1.6	0.3	
U2 - Toxicity prior Beta(0.3, 0.7)	15	7.9	4.1	2	0.5	
U3 - Toxicity prior Beta(0.3, 0.7)	15.7	7.7	3.8	1.8	0.4	
U2 - Toxicity prior Beta(0.5, 0.5)	14.3	8.1	4.4	2.1	0.5	
U3 - Toxicity prior Beta(0.5, 0.5)	14.8	8	4.1	2	0.5	
U2 - Toxicity prior Beta(1, 1)	14.2	8.1	4.4	2.2	0.5	
U3 - Toxicity prior Beta(1, 1)	15	7.9	4	2	0.5	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	1.1	3.3	1	0.6	0.1	
U3 - Toxicity prior Beta(0.15, 0.85)	1.1	3.2	1	0.5	0.1	
U2 - Toxicity prior Beta(0.3, 0.7)	1.1	3.3	1	0.6	0.1	
U3 - Toxicity prior Beta(0.3, 0.7)	1.2	3.2	0.9	0.5	0.1	
U2 - Toxicity prior Beta(0.5, 0.5)	1.1	3.2	1	0.6	0.1	
U3 - Toxicity prior Beta(0.5, 0.5)	1.2	3.1	0.9	0.5	0.1	
U2 - Toxicity prior Beta(1, 1)	1.1	2.7	0.9	0.5	0.1	
U3 - Toxicity prior Beta(1, 1)	1.2	2.6	0.9	0.5	0.1	
Scenario 12						
DLT rate	0.06	0.09	0.18	0.29	0.44	
Efficacy rate	0.05	0.1	0.12	0.4	0.6	
Utility 2	0	0	-0.1	0.1	0.2	
Utility 3	40.6	42.4	40	52.4	58.4	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	14	10.7	8.3	41.5	25.2	0.3
U3 - Toxicity prior Beta(0.15, 0.85)	19.5	11.7	7.4	39.4	21.7	0.3
U2 - Toxicity prior Beta(0.3, 0.7)	13.3	9.5	7.5	41.5	27.9	0.3
U3 - Toxicity prior Beta(0.3, 0.7)	18.1	11.5	6.6	39.6	24	0.2
U2 - Toxicity prior Beta(0.5, 0.5)	12.9	9.2	8.1	40.7	28.8	0.3
U3 - Toxicity prior Beta(0.5, 0.5)	17	11.5	6.8	38.5	25.9	0.3
U2 - Toxicity prior Beta(1, 1)	11.8	9.2	7.3	40.4	31	0.3
U3 - Toxicity prior Beta(1, 1)	16.2	12.1	5.2	37.6	28.7	0.2
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	4.4	4.8	5.3	9.3	6.2	
U3 - Toxicity prior Beta(0.15, 0.85)	4.8	4.9	5.1	9.5	5.8	
U2 - Toxicity prior Beta(0.3, 0.7)	4.2	4.6	5.2	9.3	6.6	
U3 - Toxicity prior Beta(0.3, 0.7)	4.6	4.7	5	9.6	6.1	
U2 - Toxicity prior Beta(0.5, 0.5)	4.1	4.5	5.1	9.4	7	
U3 - Toxicity prior Beta(0.5, 0.5)	4.4	4.7	4.9	9.6	6.4	
U2 - Toxicity prior Beta(1, 1)	4	4.4	4.9	9.4	7.2	
U3 - Toxicity prior Beta(1, 1)	4.3	4.5	4.7	9.7	6.7	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.3	3	1.5	2.4	1.2	
U3 - Toxicity prior Beta(0.15, 0.85)	0.4	3	1.5	2.3	1.1	
U2 - Toxicity prior Beta(0.3, 0.7)	0.3	2.9	1.5	2.3	1.2	
U3 - Toxicity prior Beta(0.3, 0.7)	0.4	3	1.4	2.3	1.1	
U2 - Toxicity prior Beta(0.5, 0.5)	0.3	2.8	1.4	2.3	1.2	
U3 - Toxicity prior Beta(0.5, 0.5)	0.3	2.9	1.4	2.2	1.1	
U2 - Toxicity prior Beta(1, 1)	0.3	2.7	1.5	2.2	1.1	
U3 - Toxicity prior Beta(1, 1)	0.3	2.7	1.4	2.2	1	

Phase I/II Operating Characteristics

Random model - Toxicity prior comparison - allocation = one_stage - IPDE alpha = 0 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 13						
DLT rate	0.21	0.29	0.3	0.39	0.46	
Efficacy rate	0.05	0.3	0.32	0.35	0.4	
Utility 2	-0.2	0	0	0	-0.1	
Utility 3	34.6	46.4	47.2	45.4	45.6	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	13.6	45.5	17.1	7.4	2.2	14.2
U3 - Toxicity prior Beta(0.15, 0.85)	17.1	45.3	15.3	5.5	1.6	15.2
U2 - Toxicity prior Beta(0.3, 0.7)	13.8	41.6	19.1	9.2	2.6	13.7
U3 - Toxicity prior Beta(0.3, 0.7)	17.2	43.7	16.5	6.6	2	14
U2 - Toxicity prior Beta(0.5, 0.5)	12.8	41.7	21.5	8.9	2.9	12.2
U3 - Toxicity prior Beta(0.5, 0.5)	16.4	43.4	19.1	6.6	1.9	12.6
U2 - Toxicity prior Beta(1, 1)	11.8	43.3	19.4	8.1	4.3	13.1
U3 - Toxicity prior Beta(1, 1)	14.9	43.7	17.7	7.6	2.4	13.7
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	8.8	9.6	5.4	3.4	1.3	
U3 - Toxicity prior Beta(0.15, 0.85)	9.1	9.9	5.2	3.2	1.1	
U2 - Toxicity prior Beta(0.3, 0.7)	8.4	9.3	5.7	3.7	1.5	
U3 - Toxicity prior Beta(0.3, 0.7)	8.6	9.7	5.5	3.5	1.3	
U2 - Toxicity prior Beta(0.5, 0.5)	7.7	9.3	6.1	4	1.7	
U3 - Toxicity prior Beta(0.5, 0.5)	8	9.7	5.9	3.7	1.4	
U2 - Toxicity prior Beta(1, 1)	7.8	9.3	5.9	3.9	1.8	
U3 - Toxicity prior Beta(1, 1)	8	9.6	5.7	3.7	1.5	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.7	4.1	1.3	1.1	0.3	
U3 - Toxicity prior Beta(0.15, 0.85)	0.7	4	1.3	1	0.2	
U2 - Toxicity prior Beta(0.3, 0.7)	0.6	3.8	1.3	1.1	0.3	
U3 - Toxicity prior Beta(0.3, 0.7)	0.7	3.8	1.3	1	0.3	
U2 - Toxicity prior Beta(0.5, 0.5)	0.5	3.5	1.3	1	0.3	
U3 - Toxicity prior Beta(0.5, 0.5)	0.6	3.5	1.3	1	0.2	
U2 - Toxicity prior Beta(1, 1)	0.6	3	1.3	0.9	0.2	
U3 - Toxicity prior Beta(1, 1)	0.6	3	1.2	0.9	0.2	
Scenario 14						
DLT rate	0.09	0.15	0.23	0.28	0.39	
Efficacy rate	0.05	0.08	0.12	0.3	0.4	
Utility 2	0	-0.1	-0.1	0	0	
Utility 3	39.4	38.8	38	46.8	48.4	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	21	15.6	13	32.4	17.4	0.6
U3 - Toxicity prior Beta(0.15, 0.85)	30.4	16.8	11.8	28.4	11.7	0.9
U2 - Toxicity prior Beta(0.3, 0.7)	20.7	16.2	11.6	32.9	18	0.6
U3 - Toxicity prior Beta(0.3, 0.7)	30.2	16.4	10.8	29.1	12.9	0.6
U2 - Toxicity prior Beta(0.5, 0.5)	18.7	15.5	12.5	33.4	19.2	0.7
U3 - Toxicity prior Beta(0.5, 0.5)	26.5	17.4	11.4	30.4	13.8	0.5
U2 - Toxicity prior Beta(1, 1)	18.3	15.5	11.7	32.4	21.4	0.7
U3 - Toxicity prior Beta(1, 1)	28.5	16.6	9.6	29.3	15.5	0.5
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	5.8	5.7	5.5	7.8	5	
U3 - Toxicity prior Beta(0.15, 0.85)	6.4	5.7	5.3	7.9	4.6	
U2 - Toxicity prior Beta(0.3, 0.7)	5.4	5.5	5.5	8	5.4	
U3 - Toxicity prior Beta(0.3, 0.7)	6	5.6	5.3	8.1	5	
U2 - Toxicity prior Beta(0.5, 0.5)	5.2	5.4	5.4	8.3	5.7	
U3 - Toxicity prior Beta(0.5, 0.5)	5.7	5.4	5.2	8.4	5.3	
U2 - Toxicity prior Beta(1, 1)	5.1	5.3	5.3	8.1	6.1	
U3 - Toxicity prior Beta(1, 1)	5.6	5.3	5.1	8.4	5.6	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.6	3.4	1.6	2.3	1	
U3 - Toxicity prior Beta(0.15, 0.85)	0.6	3.4	1.6	2.2	1	
U2 - Toxicity prior Beta(0.3, 0.7)	0.5	3.3	1.6	2.3	1	
U3 - Toxicity prior Beta(0.3, 0.7)	0.6	3.3	1.5	2.2	1	
U2 - Toxicity prior Beta(0.5, 0.5)	0.5	3.2	1.5	2.3	1	
U3 - Toxicity prior Beta(0.5, 0.5)	0.5	3.2	1.4	2.2	1	
U2 - Toxicity prior Beta(1, 1)	0.5	3	1.5	2.2	0.9	
U3 - Toxicity prior Beta(1, 1)	0.6	3	1.5	2.2	0.9	

Phase I/II Operating Characteristics

Random model - Toxicity prior comparison - allocation = one_stage - IPDE alpha = 0 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 15						
DLT rate	0.13	0.2	0.32	0.39	0.46	
Efficacy rate	0.1	0.1	0.1	0.1	0.1	
Utility 2	0	-0.1	-0.2	-0.3	-0.4	
Utility 3	40.8	38	33.2	30.4	27.6	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	51.9	26.3	13.6	5.6	1.1	1.5
U3 - Toxicity prior Beta(0.15, 0.85)	60	25.6	9.9	2.4	0.3	1.8
U2 - Toxicity prior Beta(0.3, 0.7)	53.5	26.6	11.9	5.4	1.3	1.3
U3 - Toxicity prior Beta(0.3, 0.7)	61.2	24.2	10.2	2.8	0.3	1.3
U2 - Toxicity prior Beta(0.5, 0.5)	53.5	26.8	12	5.1	1.4	1.2
U3 - Toxicity prior Beta(0.5, 0.5)	60	25.5	10.5	2.5	0.3	1.2
U2 - Toxicity prior Beta(1, 1)	50.2	28.4	12.3	6.5	1.2	1.4
U3 - Toxicity prior Beta(1, 1)	59.2	25.3	10.9	2.9	0.4	1.3
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	9.7	8	5.9	4.4	1.8	
U3 - Toxicity prior Beta(0.15, 0.85)	10.6	7.8	5.5	4.2	1.8	
U2 - Toxicity prior Beta(0.3, 0.7)	9.3	7.8	6.1	4.7	2	
U3 - Toxicity prior Beta(0.3, 0.7)	10.2	7.7	5.6	4.4	1.9	
U2 - Toxicity prior Beta(0.5, 0.5)	8.9	7.7	6.1	4.9	2.2	
U3 - Toxicity prior Beta(0.5, 0.5)	9.6	7.6	5.8	4.7	2.1	
U2 - Toxicity prior Beta(1, 1)	8.9	7.7	6.1	4.9	2.3	
U3 - Toxicity prior Beta(1, 1)	9.7	7.6	5.8	4.7	2.1	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	1.1	4	1.8	1.7	0.5	
U3 - Toxicity prior Beta(0.15, 0.85)	1.2	3.9	1.7	1.6	0.5	
U2 - Toxicity prior Beta(0.3, 0.7)	1.1	3.9	1.8	1.7	0.5	
U3 - Toxicity prior Beta(0.3, 0.7)	1.2	3.9	1.7	1.6	0.5	
U2 - Toxicity prior Beta(0.5, 0.5)	1.1	3.7	1.8	1.7	0.5	
U3 - Toxicity prior Beta(0.5, 0.5)	1.3	3.7	1.7	1.7	0.5	
U2 - Toxicity prior Beta(1, 1)	1.2	3.5	1.7	1.7	0.4	
U3 - Toxicity prior Beta(1, 1)	1.3	3.6	1.7	1.6	0.4	
Scenario 16						
DLT rate	0.02	0.06	0.09	0.14	0.26	
Efficacy rate	0.2	0.2	0.2	0.2	0.2	
Utility 2	0.2	0.1	0.1	0.1	-0.1	
Utility 3	51.2	49.6	48.4	46.4	41.6	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	43.1	19.9	17.2	12.3	7.5	0
U3 - Toxicity prior Beta(0.15, 0.85)	48.4	22	14.6	9.7	5.3	0
U2 - Toxicity prior Beta(0.3, 0.7)	43.4	21.5	15.5	12	7.6	0
U3 - Toxicity prior Beta(0.3, 0.7)	48	22.9	14.5	9.4	5.2	0
U2 - Toxicity prior Beta(0.5, 0.5)	42.6	21.7	14.9	12.8	8	0
U3 - Toxicity prior Beta(0.5, 0.5)	48.7	23.1	14.4	9.1	4.7	0
U2 - Toxicity prior Beta(1, 1)	43.1	20.7	16.9	11.6	7.7	0
U3 - Toxicity prior Beta(1, 1)	47.8	22.9	15.4	9.1	4.8	0
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	4.6	4.7	6	8.2	6.5	
U3 - Toxicity prior Beta(0.15, 0.85)	4.9	5	6.2	8.1	5.9	
U2 - Toxicity prior Beta(0.3, 0.7)	4.5	4.7	6.1	8.2	6.6	
U3 - Toxicity prior Beta(0.3, 0.7)	4.7	4.9	6.2	8.3	5.9	
U2 - Toxicity prior Beta(0.5, 0.5)	4.4	4.6	6.1	8.3	6.7	
U3 - Toxicity prior Beta(0.5, 0.5)	4.6	4.9	6.3	8.2	6	
U2 - Toxicity prior Beta(1, 1)	4.3	4.6	6.1	8.2	6.7	
U3 - Toxicity prior Beta(1, 1)	4.5	4.9	6.2	8.2	6.2	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.2	2.7	2	3	1.4	
U3 - Toxicity prior Beta(0.15, 0.85)	0.3	2.9	2	2.9	1.3	
U2 - Toxicity prior Beta(0.3, 0.7)	0.2	2.7	2	3	1.3	
U3 - Toxicity prior Beta(0.3, 0.7)	0.3	2.8	2	3	1.3	
U2 - Toxicity prior Beta(0.5, 0.5)	0.2	2.7	2	3	1.4	
U3 - Toxicity prior Beta(0.5, 0.5)	0.3	2.8	2.1	2.9	1.3	
U2 - Toxicity prior Beta(1, 1)	0.2	2.7	2	3	1.3	
U3 - Toxicity prior Beta(1, 1)	0.3	2.8	2.1	3	1.3	

Phase I/II Operating Characteristics

Random model - Toxicity prior comparison - allocation = one_stage - IPDE alpha = 0 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 17						
DLT rate	0.11	0.23	0.39	0.58	0.76	
Efficacy rate	0.3	0.5	0.5	0.5	0.5	
Utility 2	0.2	0.3	0.1	-0.1	-0.3	
Utility 3	53.6	60.8	54.4	46.8	39.6	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	22.5	56.1	18.1	3.1	0.1	0.1
U3 - Toxicity prior Beta(0.15, 0.85)	27	57.6	13.5	1.8	0	0.1
U2 - Toxicity prior Beta(0.3, 0.7)	22.7	55.1	19	3	0.1	0.1
U3 - Toxicity prior Beta(0.3, 0.7)	26	57.8	13.8	2.2	0.1	0.1
U2 - Toxicity prior Beta(0.5, 0.5)	20.5	56.8	19.2	3.2	0.1	0.2
U3 - Toxicity prior Beta(0.5, 0.5)	25.3	56.8	15.8	2	0	0.1
U2 - Toxicity prior Beta(1, 1)	20.6	53.3	22	3.7	0.3	0.1
U3 - Toxicity prior Beta(1, 1)	23.9	58	15.5	2.4	0.1	0.1
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	9	13.2	5.8	1.6	0.3	
U3 - Toxicity prior Beta(0.15, 0.85)	9.6	13.6	5.2	1.3	0.2	
U2 - Toxicity prior Beta(0.3, 0.7)	8.6	13.1	6.2	1.8	0.3	
U3 - Toxicity prior Beta(0.3, 0.7)	9.2	13.7	5.6	1.4	0.2	
U2 - Toxicity prior Beta(0.5, 0.5)	8.3	13.1	6.4	1.9	0.3	
U3 - Toxicity prior Beta(0.5, 0.5)	8.9	13.5	5.8	1.6	0.2	
U2 - Toxicity prior Beta(1, 1)	8.2	12.8	6.5	2.1	0.3	
U3 - Toxicity prior Beta(1, 1)	8.7	13.7	5.7	1.6	0.2	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.5	3.6	1.3	0.5	0.1	
U3 - Toxicity prior Beta(0.15, 0.85)	0.5	3.6	1.1	0.4	0.1	
U2 - Toxicity prior Beta(0.3, 0.7)	0.5	3.6	1.2	0.5	0.1	
U3 - Toxicity prior Beta(0.3, 0.7)	0.5	3.6	1.1	0.4	0.1	
U2 - Toxicity prior Beta(0.5, 0.5)	0.4	3.4	1.2	0.5	0.1	
U3 - Toxicity prior Beta(0.5, 0.5)	0.5	3.4	1.1	0.4	0	
U2 - Toxicity prior Beta(1, 1)	0.4	3.2	1.2	0.4	0.1	
U3 - Toxicity prior Beta(1, 1)	0.5	3.2	1	0.4	0	
Scenario 18						
DLT rate	0.02	0.04	0.1	0.2	0.35	
Efficacy rate	0.05	0.1	0.15	0.2	0.5	
Utility 2	0	0.1	0.1	0	0.1	
Utility 3	42.2	44.4	45	44	56	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	12.7	7	9.1	15.4	55.8	0
U3 - Toxicity prior Beta(0.15, 0.85)	19.7	11	9.5	11.3	48.5	0
U2 - Toxicity prior Beta(0.3, 0.7)	11.8	7.6	9	14.2	57.4	0
U3 - Toxicity prior Beta(0.3, 0.7)	19.4	9.5	9.9	11.2	50	0
U2 - Toxicity prior Beta(0.5, 0.5)	12.4	8.2	8.1	13.1	58.2	0
U3 - Toxicity prior Beta(0.5, 0.5)	19.6	11.1	8.3	10	51	0
U2 - Toxicity prior Beta(1, 1)	11.2	7.4	8.2	14.4	58.8	0
U3 - Toxicity prior Beta(1, 1)	18.5	10.3	9.1	11.7	50.4	0
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	3.5	3.9	5.1	6.5	11	
U3 - Toxicity prior Beta(0.15, 0.85)	3.7	4.1	5.1	6.6	10.5	
U2 - Toxicity prior Beta(0.3, 0.7)	3.4	3.8	5.1	6.4	11.2	
U3 - Toxicity prior Beta(0.3, 0.7)	3.6	4	5.1	6.5	10.8	
U2 - Toxicity prior Beta(0.5, 0.5)	3.4	3.8	5	6.5	11.3	
U3 - Toxicity prior Beta(0.5, 0.5)	3.6	4	5	6.5	10.9	
U2 - Toxicity prior Beta(1, 1)	3.4	3.8	4.9	6.5	11.5	
U3 - Toxicity prior Beta(1, 1)	3.5	3.9	5	6.6	11	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.1	2.7	1.5	2.5	1.4	
U3 - Toxicity prior Beta(0.15, 0.85)	0.2	2.8	1.5	2.5	1.4	
U2 - Toxicity prior Beta(0.3, 0.7)	0.1	2.7	1.5	2.5	1.5	
U3 - Toxicity prior Beta(0.3, 0.7)	0.2	2.7	1.5	2.5	1.5	
U2 - Toxicity prior Beta(0.5, 0.5)	0.1	2.7	1.5	2.5	1.5	
U3 - Toxicity prior Beta(0.5, 0.5)	0.2	2.7	1.5	2.5	1.4	
U2 - Toxicity prior Beta(1, 1)	0.1	2.6	1.5	2.5	1.4	
U3 - Toxicity prior Beta(1, 1)	0.2	2.7	1.5	2.5	1.4	

Phase I/II Operating Characteristics

Random model - Toxicity prior comparison - allocation = one_stage - IPDE alpha = 0 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 19						
DLT rate	0.11	0.26	0.39	0.6	0.77	
Efficacy rate	0.08	0.3	0.32	0.35	0.4	
Utility 2	0	0	-0.1	-0.2	-0.4	
Utility 3	40.4	47.6	43.6	37	33.2	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	17.1	58	19.7	2.7	0.1	2.4
U3 - Toxicity prior Beta(0.15, 0.85)	19.9	58.5	16.3	2.8	0	2.5
U2 - Toxicity prior Beta(0.3, 0.7)	17.1	55.1	21.8	3.6	0.1	2.3
U3 - Toxicity prior Beta(0.3, 0.7)	19.6	57.4	17.6	2.9	0	2.5
U2 - Toxicity prior Beta(0.5, 0.5)	16.1	57.6	20.3	3.7	0	2.3
U3 - Toxicity prior Beta(0.5, 0.5)	19	57.8	18.5	2.6	0	2.1
U2 - Toxicity prior Beta(1, 1)	16	56.4	21.2	4.6	0.1	1.7
U3 - Toxicity prior Beta(1, 1)	18.4	57.7	18.7	3.5	0.1	1.6
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	7.4	12	7.1	2.8	0.6	
U3 - Toxicity prior Beta(0.15, 0.85)	7.9	12.3	6.6	2.6	0.5	
U2 - Toxicity prior Beta(0.3, 0.7)	7.1	11.8	7.3	3	0.6	
U3 - Toxicity prior Beta(0.3, 0.7)	7.5	12.1	6.9	2.8	0.6	
U2 - Toxicity prior Beta(0.5, 0.5)	6.9	11.7	7.4	3.1	0.7	
U3 - Toxicity prior Beta(0.5, 0.5)	7.3	12	7	2.9	0.7	
U2 - Toxicity prior Beta(1, 1)	6.8	11.6	7.4	3.4	0.8	
U3 - Toxicity prior Beta(1, 1)	7.2	11.9	7	3.1	0.7	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.7	4.5	1.7	0.9	0.1	
U3 - Toxicity prior Beta(0.15, 0.85)	0.8	4.4	1.6	0.9	0.1	
U2 - Toxicity prior Beta(0.3, 0.7)	0.7	4.3	1.7	0.9	0.1	
U3 - Toxicity prior Beta(0.3, 0.7)	0.8	4.3	1.5	0.9	0.1	
U2 - Toxicity prior Beta(0.5, 0.5)	0.6	4	1.6	0.9	0.1	
U3 - Toxicity prior Beta(0.5, 0.5)	0.7	4	1.5	0.8	0.1	
U2 - Toxicity prior Beta(1, 1)	0.7	3.7	1.4	0.8	0.1	
U3 - Toxicity prior Beta(1, 1)	0.7	3.7	1.4	0.8	0.1	
Scenario 20						
DLT rate	0.07	0.14	0.31	0.41	0.52	
Efficacy rate	0.05	0.25	0.25	0.25	0.25	
Utility 2	0	0.1	-0.1	-0.2	-0.3	
Utility 3	40.2	49.4	42.6	38.6	34.2	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	12.3	43.1	28.8	13.4	2	0.4
U3 - Toxicity prior Beta(0.15, 0.85)	17.8	47.1	23	10.2	1.5	0.4
U2 - Toxicity prior Beta(0.3, 0.7)	12.2	42	28.9	14	2.5	0.4
U3 - Toxicity prior Beta(0.3, 0.7)	17.9	46.9	24	9.3	1.5	0.4
U2 - Toxicity prior Beta(0.5, 0.5)	12.7	42.8	29.2	12.3	2.7	0.3
U3 - Toxicity prior Beta(0.5, 0.5)	18.2	46.7	24.4	8.3	2.1	0.3
U2 - Toxicity prior Beta(1, 1)	12	39.5	29.5	14.9	3.5	0.6
U3 - Toxicity prior Beta(1, 1)	18.5	46	23.7	8.7	2.6	0.5
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	4.7	9.2	8.2	5.7	2.2	
U3 - Toxicity prior Beta(0.15, 0.85)	5.2	9.6	7.7	5.4	2	
U2 - Toxicity prior Beta(0.3, 0.7)	4.5	8.8	8.2	6	2.5	
U3 - Toxicity prior Beta(0.3, 0.7)	4.9	9.5	7.8	5.6	2.1	
U2 - Toxicity prior Beta(0.5, 0.5)	4.3	8.7	8.1	6.1	2.7	
U3 - Toxicity prior Beta(0.5, 0.5)	4.7	9.1	7.8	5.9	2.4	
U2 - Toxicity prior Beta(1, 1)	4.3	8.4	8	6.2	2.9	
U3 - Toxicity prior Beta(1, 1)	4.7	9	7.7	6	2.5	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.4	3.7	2.3	1.8	0.6	
U3 - Toxicity prior Beta(0.15, 0.85)	0.5	3.7	2.2	1.7	0.6	
U2 - Toxicity prior Beta(0.3, 0.7)	0.4	3.6	2.2	1.8	0.6	
U3 - Toxicity prior Beta(0.3, 0.7)	0.5	3.6	2.1	1.7	0.6	
U2 - Toxicity prior Beta(0.5, 0.5)	0.4	3.5	2.1	1.9	0.6	
U3 - Toxicity prior Beta(0.5, 0.5)	0.5	3.5	2.1	1.8	0.5	
U2 - Toxicity prior Beta(1, 1)	0.4	3.3	2.1	1.8	0.5	
U3 - Toxicity prior Beta(1, 1)	0.5	3.3	2.1	1.7	0.5	

Phase I/II Operating Characteristics

Random model - Toxicity prior comparison - allocation = one_stage - IPDE alpha = 0 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 21						
DLT rate	0.37	0.56	0.69	0.83	0.88	
Efficacy rate	0.5	0.5	0.5	0.5	0.5	
Utility 2	0.1	-0.1	-0.2	-0.3	-0.4	
Utility 3	55.2	47.6	42.4	36.8	34.8	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	57.9	1.8	0	0	0	40.3
U3 - Toxicity prior Beta(0.15, 0.85)	58.4	1.4	0	0	0	40.2
U2 - Toxicity prior Beta(0.3, 0.7)	58.4	2.2	0.1	0	0	39.3
U3 - Toxicity prior Beta(0.3, 0.7)	59.2	1.6	0	0	0	39.2
U2 - Toxicity prior Beta(0.5, 0.5)	58.2	2.2	0	0	0	39.6
U3 - Toxicity prior Beta(0.5, 0.5)	58.5	1.8	0	0	0	39.7
U2 - Toxicity prior Beta(1, 1)	58.4	2.3	0	0	0	39.3
U3 - Toxicity prior Beta(1, 1)	58.5	1.9	0	0	0	39.6
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	20.1	2.7	0.3	0	0	
U3 - Toxicity prior Beta(0.15, 0.85)	20.2	2.6	0.3	0.1	0	
U2 - Toxicity prior Beta(0.3, 0.7)	20	2.9	0.4	0.1	0	
U3 - Toxicity prior Beta(0.3, 0.7)	20.1	2.8	0.4	0	0	
U2 - Toxicity prior Beta(0.5, 0.5)	19.8	3	0.5	0.1	0	
U3 - Toxicity prior Beta(0.5, 0.5)	19.9	2.9	0.4	0.1	0	
U2 - Toxicity prior Beta(1, 1)	19.8	3	0.5	0.1	0	
U3 - Toxicity prior Beta(1, 1)	20	2.9	0.4	0.1	0	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.2	1.1	0.1	0	0	
U3 - Toxicity prior Beta(0.15, 0.85)	0.2	1	0.1	0	0	
U2 - Toxicity prior Beta(0.3, 0.7)	0.3	1	0.1	0	0	
U3 - Toxicity prior Beta(0.3, 0.7)	0.3	1	0.1	0	0	
U2 - Toxicity prior Beta(0.5, 0.5)	0.2	1	0.1	0	0	
U3 - Toxicity prior Beta(0.5, 0.5)	0.2	0.9	0.1	0	0	
U2 - Toxicity prior Beta(1, 1)	0.1	0.7	0.1	0	0	
U3 - Toxicity prior Beta(1, 1)	0.2	0.7	0.1	0	0	
Scenario 22						
DLT rate	0.07	0.27	0.5	0.56	0.78	
Efficacy rate	0.08	0.25	0.25	0.5	0.5	
Utility 2	0	0	-0.2	-0.1	-0.3	
Utility 3	42	44.2	35	47.6	38.8	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	20.5	59	14.7	4.2	0.1	1.5
U3 - Toxicity prior Beta(0.15, 0.85)	23.4	58.4	13.1	3.7	0	1.4
U2 - Toxicity prior Beta(0.3, 0.7)	20.7	57.4	16	4.3	0.1	1.5
U3 - Toxicity prior Beta(0.3, 0.7)	23.8	56.8	13.6	4.4	0	1.4
U2 - Toxicity prior Beta(0.5, 0.5)	19.9	58	15.9	5	0.1	1.1
U3 - Toxicity prior Beta(0.5, 0.5)	25.2	56.9	12.7	4.2	0	1
U2 - Toxicity prior Beta(1, 1)	20.1	57.8	15.8	4.8	0.2	1.3
U3 - Toxicity prior Beta(1, 1)	23.9	56	13.7	5.2	0	1.2
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	8.2	12.4	6.2	2.7	0.3	
U3 - Toxicity prior Beta(0.15, 0.85)	8.7	12.5	5.8	2.6	0.3	
U2 - Toxicity prior Beta(0.3, 0.7)	7.9	12.2	6.4	3	0.4	
U3 - Toxicity prior Beta(0.3, 0.7)	8.4	12.3	6.1	2.8	0.4	
U2 - Toxicity prior Beta(0.5, 0.5)	7.7	12.1	6.5	3.2	0.5	
U3 - Toxicity prior Beta(0.5, 0.5)	8.2	12.1	6.2	3	0.5	
U2 - Toxicity prior Beta(1, 1)	7.6	12	6.5	3.3	0.6	
U3 - Toxicity prior Beta(1, 1)	8.1	12	6.2	3.2	0.5	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	1	4.5	1.7	0.8	0.1	
U3 - Toxicity prior Beta(0.15, 0.85)	1	4.4	1.7	0.7	0.1	
U2 - Toxicity prior Beta(0.3, 0.7)	1	4.4	1.6	0.7	0.1	
U3 - Toxicity prior Beta(0.3, 0.7)	1.1	4.3	1.6	0.7	0.1	
U2 - Toxicity prior Beta(0.5, 0.5)	1	4.2	1.5	0.7	0.1	
U3 - Toxicity prior Beta(0.5, 0.5)	1	4.1	1.4	0.6	0.1	
U2 - Toxicity prior Beta(1, 1)	0.9	3.8	1.3	0.7	0.1	
U3 - Toxicity prior Beta(1, 1)	1	3.8	1.3	0.6	0.1	

Phase I/II Operating Characteristics

Random model - Toxicity prior comparison - allocation = one_stage - IPDE alpha = 0 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 23						
DLT rate	0.32	0.55	0.8	0.94	0.97	
Efficacy rate	0.05	0.05	0.05	0.05	0.05	
Utility 2	-0.3	-0.5	-0.8	-0.9	-0.9	
Utility 3	30.2	21	11	5.4	4.2	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	22.6	5.7	0.2	0	0	71.5
U3 - Toxicity prior Beta(0.15, 0.85)	22.2	6.5	0.3	0	0	71
U2 - Toxicity prior Beta(0.3, 0.7)	23.4	4.8	0.4	0	0	71.4
U3 - Toxicity prior Beta(0.3, 0.7)	22.5	6.1	0.2	0	0	71.2
U2 - Toxicity prior Beta(0.5, 0.5)	24.2	5	0	0	0	70.8
U3 - Toxicity prior Beta(0.5, 0.5)	22.6	5.8	0.3	0	0	71.3
U2 - Toxicity prior Beta(1, 1)	22.7	4.4	0.2	0	0	72.7
U3 - Toxicity prior Beta(1, 1)	22.4	5.2	0.3	0	0	72.1
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	15.4	5.3	1.1	0.2	0	
U3 - Toxicity prior Beta(0.15, 0.85)	15.5	5.1	1.1	0.2	0	
U2 - Toxicity prior Beta(0.3, 0.7)	15.4	5.3	1.3	0.2	0	
U3 - Toxicity prior Beta(0.3, 0.7)	15.5	5.2	1.3	0.2	0	
U2 - Toxicity prior Beta(0.5, 0.5)	15.3	5.4	1.5	0.2	0	
U3 - Toxicity prior Beta(0.5, 0.5)	15.4	5.2	1.6	0.3	0	
U2 - Toxicity prior Beta(1, 1)	15.1	5.2	1.5	0.3	0	
U3 - Toxicity prior Beta(1, 1)	15.3	5.1	1.5	0.3	0	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.8	2.7	0.3	0.1	0	
U3 - Toxicity prior Beta(0.15, 0.85)	0.8	2.6	0.3	0.1	0	
U2 - Toxicity prior Beta(0.3, 0.7)	0.8	2.5	0.3	0	0	
U3 - Toxicity prior Beta(0.3, 0.7)	0.8	2.4	0.3	0	0	
U2 - Toxicity prior Beta(0.5, 0.5)	0.6	2.2	0.3	0	0	
U3 - Toxicity prior Beta(0.5, 0.5)	0.6	2.1	0.3	0	0	
U2 - Toxicity prior Beta(1, 1)	0.5	1.7	0.2	0	0	
U3 - Toxicity prior Beta(1, 1)	0.5	1.6	0.2	0	0	
Scenario 24						
DLT rate	0.01	0.02	0.05	0.1	0.25	
Efficacy rate	0.05	0.08	0.2	0.5	0.5	
Utility 2	0	0.1	0.1	0.4	0.2	
Utility 3	42.6	44	50	66	60	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	2.4	1.2	4.4	50.5	41.5	0
U3 - Toxicity prior Beta(0.15, 0.85)	4.6	1.9	5.3	54.1	34.1	0
U2 - Toxicity prior Beta(0.3, 0.7)	2.5	1.4	4.2	50.2	41.7	0
U3 - Toxicity prior Beta(0.3, 0.7)	4.1	1.8	4.9	53.6	35.6	0
U2 - Toxicity prior Beta(0.5, 0.5)	2.3	0.8	3.9	51	42	0
U3 - Toxicity prior Beta(0.5, 0.5)	4.3	2	4.9	54	34.8	0
U2 - Toxicity prior Beta(1, 1)	2.7	0.8	3.9	51.3	41.3	0
U3 - Toxicity prior Beta(1, 1)	4.4	1.7	5.7	52.8	35.4	0
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	3.1	3.1	3.9	11.7	8.2	
U3 - Toxicity prior Beta(0.15, 0.85)	3.1	3.2	4	12.2	7.5	
U2 - Toxicity prior Beta(0.3, 0.7)	3.1	3.1	3.9	11.6	8.2	
U3 - Toxicity prior Beta(0.3, 0.7)	3.1	3.2	3.9	12.2	7.6	
U2 - Toxicity prior Beta(0.5, 0.5)	3.1	3.1	3.9	11.7	8.2	
U3 - Toxicity prior Beta(0.5, 0.5)	3.1	3.2	4	12.2	7.6	
U2 - Toxicity prior Beta(1, 1)	3.1	3.1	3.9	11.6	8.3	
U3 - Toxicity prior Beta(1, 1)	3.1	3.1	4	12.2	7.6	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0	2.4	1	2.5	1.4	
U3 - Toxicity prior Beta(0.15, 0.85)	0	2.4	1.1	2.5	1.3	
U2 - Toxicity prior Beta(0.3, 0.7)	0	2.4	1	2.4	1.4	
U3 - Toxicity prior Beta(0.3, 0.7)	0	2.4	1.1	2.5	1.3	
U2 - Toxicity prior Beta(0.5, 0.5)	0	2.4	1.1	2.5	1.4	
U3 - Toxicity prior Beta(0.5, 0.5)	0	2.4	1.1	2.5	1.3	
U2 - Toxicity prior Beta(1, 1)	0	2.4	1.1	2.5	1.4	
U3 - Toxicity prior Beta(1, 1)	0	2.4	1.1	2.5	1.3	

Phase I/II Operating Characteristics

Random model - Toxicity prior comparison - allocation = one_stage - IPDE alpha = 0 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 25						
DLT rate	0.09	0.18	0.37	0.63	0.77	
Efficacy rate	0.2	0.25	0.4	0.45	0.5	
Utility 2	0.1	0.1	0	-0.2	-0.3	
Utility 3	48.4	47.8	49.2	41.8	39.2	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	31.3	27.3	34.3	6.7	0.3	0.1
U3 - Toxicity prior Beta(0.15, 0.85)	37.9	26.6	30.1	5.1	0.2	0.1
U2 - Toxicity prior Beta(0.3, 0.7)	29.9	27.8	34.5	7.5	0.2	0.1
U3 - Toxicity prior Beta(0.3, 0.7)	35.7	27.6	29.8	6.5	0.3	0.1
U2 - Toxicity prior Beta(0.5, 0.5)	28.8	28	36.2	6.7	0.2	0.1
U3 - Toxicity prior Beta(0.5, 0.5)	36.7	26.4	30.8	5.7	0.3	0.1
U2 - Toxicity prior Beta(1, 1)	31.4	26.6	34.5	7.1	0.3	0.1
U3 - Toxicity prior Beta(1, 1)	37	26.2	30.1	6.2	0.3	0.2
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	8.7	8.7	8.9	3.2	0.5	
U3 - Toxicity prior Beta(0.15, 0.85)	9.3	8.7	8.6	3	0.4	
U2 - Toxicity prior Beta(0.3, 0.7)	8.4	8.5	9.1	3.4	0.5	
U3 - Toxicity prior Beta(0.3, 0.7)	8.9	8.7	8.8	3.1	0.5	
U2 - Toxicity prior Beta(0.5, 0.5)	8	8.5	9.3	3.6	0.7	
U3 - Toxicity prior Beta(0.5, 0.5)	8.5	8.7	8.9	3.3	0.6	
U2 - Toxicity prior Beta(1, 1)	8	8.3	9.3	3.7	0.7	
U3 - Toxicity prior Beta(1, 1)	8.4	8.4	9.1	3.5	0.6	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.7	3.6	1.9	1	0.1	
U3 - Toxicity prior Beta(0.15, 0.85)	0.8	3.6	1.8	1	0.1	
U2 - Toxicity prior Beta(0.3, 0.7)	0.8	3.6	1.8	1.1	0.1	
U3 - Toxicity prior Beta(0.3, 0.7)	0.8	3.5	1.7	1	0.1	
U2 - Toxicity prior Beta(0.5, 0.5)	0.7	3.5	1.7	1	0.1	
U3 - Toxicity prior Beta(0.5, 0.5)	0.8	3.4	1.7	1	0.1	
U2 - Toxicity prior Beta(1, 1)	0.8	3.3	1.6	1	0.1	
U3 - Toxicity prior Beta(1, 1)	0.9	3.2	1.6	0.9	0.1	
Scenario 26						
DLT rate	0.15	0.22	0.3	0.62	0.8	
Efficacy rate	0.1	0.2	0.35	0.45	0.55	
Utility 2	-0.1	0	0.1	-0.2	-0.2	
Utility 3	40	43.2	49	42.2	41	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	26.7	29.9	33.4	6.9	0.1	3
U3 - Toxicity prior Beta(0.15, 0.85)	30.9	30.6	29.3	6.2	0	3
U2 - Toxicity prior Beta(0.3, 0.7)	25.1	28.8	35.1	8.2	0.1	2.7
U3 - Toxicity prior Beta(0.3, 0.7)	29.5	29.5	30.3	7.8	0.1	2.8
U2 - Toxicity prior Beta(0.5, 0.5)	24.7	29.9	35.5	7.5	0.1	2.3
U3 - Toxicity prior Beta(0.5, 0.5)	29.2	29.5	31.3	7.6	0	2.4
U2 - Toxicity prior Beta(1, 1)	26.4	28.5	34.3	8.5	0.1	2.2
U3 - Toxicity prior Beta(1, 1)	29.7	28.8	31.6	7.2	0.2	2.5
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	8.3	8.6	8.6	3.6	0.6	
U3 - Toxicity prior Beta(0.15, 0.85)	8.9	8.7	8.3	3.4	0.5	
U2 - Toxicity prior Beta(0.3, 0.7)	7.9	8.4	8.9	3.9	0.7	
U3 - Toxicity prior Beta(0.3, 0.7)	8.5	8.5	8.6	3.6	0.6	
U2 - Toxicity prior Beta(0.5, 0.5)	7.5	8.3	9.1	4.1	0.8	
U3 - Toxicity prior Beta(0.5, 0.5)	8	8.5	8.8	3.8	0.7	
U2 - Toxicity prior Beta(1, 1)	7.6	8.2	9	4.2	0.8	
U3 - Toxicity prior Beta(1, 1)	8.1	8.3	8.7	3.9	0.7	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.7	3.9	1.7	1.3	0.1	
U3 - Toxicity prior Beta(0.15, 0.85)	0.8	3.8	1.6	1.2	0.1	
U2 - Toxicity prior Beta(0.3, 0.7)	0.7	3.8	1.6	1.3	0.1	
U3 - Toxicity prior Beta(0.3, 0.7)	0.8	3.8	1.5	1.2	0.1	
U2 - Toxicity prior Beta(0.5, 0.5)	0.7	3.6	1.6	1.3	0.1	
U3 - Toxicity prior Beta(0.5, 0.5)	0.8	3.5	1.5	1.2	0.1	
U2 - Toxicity prior Beta(1, 1)	0.7	3.3	1.5	1.1	0.1	
U3 - Toxicity prior Beta(1, 1)	0.9	3.2	1.4	1.1	0.1	

Phase I/II Operating Characteristics

Random model - Toxicity prior comparison - allocation = one_stage - IPDE alpha = 0 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 27						
DLT rate	0.11	0.17	0.3	0.5	0.62	
Efficacy rate	0.1	0.2	0.4	0.42	0.45	
Utility 2	0	0	0.1	-0.1	-0.2	
Utility 3	41.6	45.2	52	45.2	42.2	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	16.8	26.6	45.9	9.2	0.8	0.7
U3 - Toxicity prior Beta(0.15, 0.85)	23.1	26.3	40.3	9	0.6	0.7
U2 - Toxicity prior Beta(0.3, 0.7)	16.9	24	46.1	11.8	0.7	0.5
U3 - Toxicity prior Beta(0.3, 0.7)	22.9	24.4	41.3	10.4	0.5	0.5
U2 - Toxicity prior Beta(0.5, 0.5)	16.3	22.6	47	12.7	0.9	0.5
U3 - Toxicity prior Beta(0.5, 0.5)	20.8	24.6	43.3	10.2	0.6	0.5
U2 - Toxicity prior Beta(1, 1)	17.8	22.2	45.2	12.9	1.4	0.5
U3 - Toxicity prior Beta(1, 1)	22.7	23.9	40.9	11.4	0.7	0.4
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	6.3	7.7	10.2	4.6	1.2	
U3 - Toxicity prior Beta(0.15, 0.85)	6.8	7.8	10.1	4.3	1	
U2 - Toxicity prior Beta(0.3, 0.7)	6	7.5	10.3	4.9	1.3	
U3 - Toxicity prior Beta(0.3, 0.7)	6.6	7.7	10.1	4.5	1.1	
U2 - Toxicity prior Beta(0.5, 0.5)	5.7	7.4	10.4	5	1.4	
U3 - Toxicity prior Beta(0.5, 0.5)	6.1	7.6	10.3	4.7	1.2	
U2 - Toxicity prior Beta(1, 1)	5.8	7.1	10.2	5.3	1.6	
U3 - Toxicity prior Beta(1, 1)	6.2	7.4	10.2	4.9	1.3	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.5	3.6	2	1.5	0.3	
U3 - Toxicity prior Beta(0.15, 0.85)	0.6	3.6	2	1.4	0.2	
U2 - Toxicity prior Beta(0.3, 0.7)	0.5	3.5	2	1.5	0.3	
U3 - Toxicity prior Beta(0.3, 0.7)	0.6	3.5	1.9	1.4	0.2	
U2 - Toxicity prior Beta(0.5, 0.5)	0.5	3.3	1.9	1.5	0.3	
U3 - Toxicity prior Beta(0.5, 0.5)	0.6	3.3	1.9	1.4	0.2	
U2 - Toxicity prior Beta(1, 1)	0.5	3.1	1.8	1.4	0.2	
U3 - Toxicity prior Beta(1, 1)	0.6	3.2	1.8	1.3	0.2	
Scenario 28						
DLT rate	0.2	0.3	0.5	0.6	0.77	
Efficacy rate	0.05	0.3	0.32	0.35	0.4	
Utility 2	-0.1	0	-0.2	-0.2	-0.4	
Utility 3	35	46	39.2	37	33.2	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	21.4	51.4	9.1	0.9	0	17.2
U3 - Toxicity prior Beta(0.15, 0.85)	22	51.8	8.3	0.5	0	17.4
U2 - Toxicity prior Beta(0.3, 0.7)	20.5	53.1	9.9	0.6	0	15.9
U3 - Toxicity prior Beta(0.3, 0.7)	22.4	51.6	8.8	0.5	0	16.7
U2 - Toxicity prior Beta(0.5, 0.5)	19.1	54.8	10.3	0.3	0	15.5
U3 - Toxicity prior Beta(0.5, 0.5)	21.7	54	8	0.6	0.1	15.6
U2 - Toxicity prior Beta(1, 1)	18.5	53.3	10.6	0.9	0	16.7
U3 - Toxicity prior Beta(1, 1)	19.5	54.1	9	0.8	0	16.6
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	10.3	12	4.5	1.5	0.2	
U3 - Toxicity prior Beta(0.15, 0.85)	10.8	11.8	4.3	1.4	0.2	
U2 - Toxicity prior Beta(0.3, 0.7)	10	12	4.8	1.7	0.3	
U3 - Toxicity prior Beta(0.3, 0.7)	10.3	11.9	4.6	1.6	0.3	
U2 - Toxicity prior Beta(0.5, 0.5)	9.4	12	5.1	1.8	0.4	
U3 - Toxicity prior Beta(0.5, 0.5)	9.8	12.2	4.8	1.7	0.3	
U2 - Toxicity prior Beta(1, 1)	9.4	11.8	5.1	1.9	0.4	
U3 - Toxicity prior Beta(1, 1)	9.8	11.9	4.8	1.8	0.3	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	1	4.4	1.3	0.5	0.1	
U3 - Toxicity prior Beta(0.15, 0.85)	1	4.3	1.2	0.5	0.1	
U2 - Toxicity prior Beta(0.3, 0.7)	0.9	4.3	1.2	0.5	0.1	
U3 - Toxicity prior Beta(0.3, 0.7)	1	4.3	1.2	0.4	0.1	
U2 - Toxicity prior Beta(0.5, 0.5)	0.9	3.9	1.2	0.5	0.1	
U3 - Toxicity prior Beta(0.5, 0.5)	0.9	3.8	1.1	0.4	0.1	
U2 - Toxicity prior Beta(1, 1)	0.8	3.3	1.1	0.4	0	
U3 - Toxicity prior Beta(1, 1)	0.9	3.3	1	0.4	0	

Phase I/II Operating Characteristics

Random model - Toxicity prior comparison - allocation = one_stage - IPDE alpha = 0 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 29						
DLT rate	0.19	0.3	0.56	0.74	0.86	
Efficacy rate	0.4	0.4	0.4	0.4	0.4	
Utility 2	0.2	0.1	-0.2	-0.3	-0.5	
Utility 3	56.4	52	41.6	34.4	29.6	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	65.7	26.9	4.9	0.2	0	2.3
U3 - Toxicity prior Beta(0.15, 0.85)	70	24.5	3.2	0.1	0	2.2
U2 - Toxicity prior Beta(0.3, 0.7)	63.4	28.3	5.6	0.4	0	2.3
U3 - Toxicity prior Beta(0.3, 0.7)	68.9	24.9	3.8	0.2	0	2.2
U2 - Toxicity prior Beta(0.5, 0.5)	63.5	28.2	5.7	0.3	0	2.3
U3 - Toxicity prior Beta(0.5, 0.5)	67.3	26.2	3.9	0.4	0	2.2
U2 - Toxicity prior Beta(1, 1)	62	29.9	5.5	0.3	0	2.3
U3 - Toxicity prior Beta(1, 1)	64.6	29.6	3.3	0.3	0	2.2
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	17.2	8.6	2.9	0.7	0.1	
U3 - Toxicity prior Beta(0.15, 0.85)	18.2	8.2	2.5	0.6	0.1	
U2 - Toxicity prior Beta(0.3, 0.7)	16.8	8.6	3.2	0.8	0.1	
U3 - Toxicity prior Beta(0.3, 0.7)	17.6	8.5	2.7	0.7	0.1	
U2 - Toxicity prior Beta(0.5, 0.5)	16.2	9	3.4	0.9	0.1	
U3 - Toxicity prior Beta(0.5, 0.5)	17	8.8	2.9	0.7	0.1	
U2 - Toxicity prior Beta(1, 1)	16.1	9	3.4	0.9	0.1	
U3 - Toxicity prior Beta(1, 1)	16.9	8.8	3	0.8	0.1	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.8	3	0.9	0.2	0	
U3 - Toxicity prior Beta(0.15, 0.85)	0.8	2.9	0.7	0.2	0	
U2 - Toxicity prior Beta(0.3, 0.7)	0.8	3	0.9	0.2	0	
U3 - Toxicity prior Beta(0.3, 0.7)	0.8	2.9	0.7	0.2	0	
U2 - Toxicity prior Beta(0.5, 0.5)	0.8	2.9	0.8	0.2	0	
U3 - Toxicity prior Beta(0.5, 0.5)	0.8	2.7	0.7	0.2	0	
U2 - Toxicity prior Beta(1, 1)	0.8	2.5	0.7	0.2	0	
U3 - Toxicity prior Beta(1, 1)	0.8	2.4	0.6	0.1	0	
Scenario 30						
DLT rate	0.05	0.08	0.17	0.27	0.54	
Efficacy rate	0.12	0.15	0.2	0.4	0.5	
Utility 2	0.1	0.1	0	0.1	0	
Utility 3	45.2	45.8	45.2	53.2	48.4	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	19	11.1	13.3	40.5	16.1	0
U3 - Toxicity prior Beta(0.15, 0.85)	28.1	12.4	12.1	37.3	10.1	0
U2 - Toxicity prior Beta(0.3, 0.7)	19.7	10.3	12.3	41.6	16.1	0
U3 - Toxicity prior Beta(0.3, 0.7)	26.9	12.3	11.4	37.1	12.3	0
U2 - Toxicity prior Beta(0.5, 0.5)	20.1	10.8	11.9	40.1	17.1	0
U3 - Toxicity prior Beta(0.5, 0.5)	27.3	12	10.4	36.4	13.9	0
U2 - Toxicity prior Beta(1, 1)	19.4	9.8	11.9	41.1	17.8	0
U3 - Toxicity prior Beta(1, 1)	27.5	11.5	10.7	37.3	13	0
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	4.3	4.6	5.9	10	5.3	
U3 - Toxicity prior Beta(0.15, 0.85)	4.5	4.7	5.9	10.3	4.7	
U2 - Toxicity prior Beta(0.3, 0.7)	4.1	4.5	5.8	10.1	5.4	
U3 - Toxicity prior Beta(0.3, 0.7)	4.3	4.6	5.7	10.3	5	
U2 - Toxicity prior Beta(0.5, 0.5)	4	4.4	5.8	10.1	5.8	
U3 - Toxicity prior Beta(0.5, 0.5)	4.2	4.6	5.7	10.3	5.2	
U2 - Toxicity prior Beta(1, 1)	4	4.3	5.6	10.2	6	
U3 - Toxicity prior Beta(1, 1)	4.2	4.5	5.6	10.4	5.4	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.2	2.7	1.7	2.4	1.3	
U3 - Toxicity prior Beta(0.15, 0.85)	0.3	2.8	1.7	2.3	1.2	
U2 - Toxicity prior Beta(0.3, 0.7)	0.2	2.7	1.6	2.4	1.3	
U3 - Toxicity prior Beta(0.3, 0.7)	0.3	2.7	1.6	2.4	1.2	
U2 - Toxicity prior Beta(0.5, 0.5)	0.2	2.7	1.6	2.4	1.3	
U3 - Toxicity prior Beta(0.5, 0.5)	0.2	2.7	1.6	2.3	1.2	
U2 - Toxicity prior Beta(1, 1)	0.2	2.6	1.6	2.3	1.2	
U3 - Toxicity prior Beta(1, 1)	0.3	2.6	1.6	2.3	1.1	

Phase I/II Operating Characteristics

Random model - Toxicity prior comparison - allocation = one_stage - IPDE alpha = 0 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 31						
DLT rate	0.04	0.06	0.1	0.2	0.6	
Efficacy rate	0.2	0.5	0.5	0.5	0.5	
Utility 2	0.2	0.4	0.4	0.3	-0.1	
Utility 3	50.4	67.6	66	62	46	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	5.5	31.7	27.7	28	7.1	0
U3 - Toxicity prior Beta(0.15, 0.85)	8.1	37.4	27	22.6	4.9	0
U2 - Toxicity prior Beta(0.3, 0.7)	6.4	31.4	26.8	28.4	7	0
U3 - Toxicity prior Beta(0.3, 0.7)	7.7	37.1	25.8	24.1	5.3	0
U2 - Toxicity prior Beta(0.5, 0.5)	6.2	31.4	26.5	28	7.9	0
U3 - Toxicity prior Beta(0.5, 0.5)	8.4	36.2	27.1	23.1	5.2	0
U2 - Toxicity prior Beta(1, 1)	6	31.5	27	28.4	7.1	0
U3 - Toxicity prior Beta(1, 1)	8.7	37.1	26.1	23.3	4.8	0
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	3.8	9.7	7.5	6.4	2.6	
U3 - Toxicity prior Beta(0.15, 0.85)	3.9	10.7	7.5	5.9	2	
U2 - Toxicity prior Beta(0.3, 0.7)	3.8	9.9	7.4	6.3	2.5	
U3 - Toxicity prior Beta(0.3, 0.7)	3.9	10.7	7.3	6	2.1	
U2 - Toxicity prior Beta(0.5, 0.5)	3.8	9.7	7.3	6.4	2.7	
U3 - Toxicity prior Beta(0.5, 0.5)	3.8	10.6	7.5	6	2.1	
U2 - Toxicity prior Beta(1, 1)	3.8	9.7	7.3	6.4	2.8	
U3 - Toxicity prior Beta(1, 1)	3.9	10.7	7.4	5.9	2.2	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.1	2.6	1.5	1.4	0.7	
U3 - Toxicity prior Beta(0.15, 0.85)	0.1	2.7	1.4	1.2	0.6	
U2 - Toxicity prior Beta(0.3, 0.7)	0.1	2.6	1.4	1.4	0.7	
U3 - Toxicity prior Beta(0.3, 0.7)	0.1	2.7	1.4	1.3	0.6	
U2 - Toxicity prior Beta(0.5, 0.5)	0.1	2.5	1.4	1.4	0.7	
U3 - Toxicity prior Beta(0.5, 0.5)	0.1	2.6	1.4	1.2	0.6	
U2 - Toxicity prior Beta(1, 1)	0	2.5	1.5	1.4	0.7	
U3 - Toxicity prior Beta(1, 1)	0.1	2.6	1.4	1.2	0.6	
Scenario 32						
DLT rate	0.02	0.05	0.11	0.3	0.36	
Efficacy rate	0.08	0.15	0.3	0.3	0.3	
Utility 2	0.1	0.1	0.2	0	-0.1	
Utility 3	44	47	53.6	46	43.6	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	15.1	12.9	33.7	24.7	13.6	0
U3 - Toxicity prior Beta(0.15, 0.85)	21	15.2	33.9	21.7	8.2	0
U2 - Toxicity prior Beta(0.3, 0.7)	14.9	13	33.9	25.5	12.7	0
U3 - Toxicity prior Beta(0.3, 0.7)	20.9	15.1	34.2	21.6	8.2	0
U2 - Toxicity prior Beta(0.5, 0.5)	14.5	12.3	34.2	24.5	14.5	0
U3 - Toxicity prior Beta(0.5, 0.5)	20.3	15.1	33.9	21.5	9.2	0
U2 - Toxicity prior Beta(1, 1)	14.7	12.3	33.5	23.8	15.7	0
U3 - Toxicity prior Beta(1, 1)	20.9	15.5	33.6	21.2	8.8	0
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	3.5	4	8.3	8.4	5.8	
U3 - Toxicity prior Beta(0.15, 0.85)	3.6	4.4	8.6	8.3	5.1	
U2 - Toxicity prior Beta(0.3, 0.7)	3.5	4	8.3	8.4	5.9	
U3 - Toxicity prior Beta(0.3, 0.7)	3.6	4.3	8.5	8.4	5.3	
U2 - Toxicity prior Beta(0.5, 0.5)	3.4	3.9	8.1	8.4	6.2	
U3 - Toxicity prior Beta(0.5, 0.5)	3.6	4.2	8.4	8.5	5.4	
U2 - Toxicity prior Beta(1, 1)	3.4	3.9	8	8.5	6.3	
U3 - Toxicity prior Beta(1, 1)	3.5	4.1	8.2	8.5	5.6	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.1	2.7	1.8	2.8	1.1	
U3 - Toxicity prior Beta(0.15, 0.85)	0.1	2.8	1.9	2.7	1	
U2 - Toxicity prior Beta(0.3, 0.7)	0.1	2.7	1.8	2.8	1.1	
U3 - Toxicity prior Beta(0.3, 0.7)	0.1	2.7	1.8	2.8	1	
U2 - Toxicity prior Beta(0.5, 0.5)	0.1	2.6	1.8	2.8	1.2	
U3 - Toxicity prior Beta(0.5, 0.5)	0.1	2.7	1.8	2.8	1.1	
U2 - Toxicity prior Beta(1, 1)	0.1	2.6	1.7	2.8	1.1	
U3 - Toxicity prior Beta(1, 1)	0.1	2.7	1.8	2.8	1	

Phase I/II Operating Characteristics

Random model - Toxicity prior comparison - allocation = one_stage - IPDE alpha = 0 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 33						
DLT rate	0.01	0.02	0.09	0.32	0.38	
Efficacy rate	0.05	0.08	0.2	0.4	0.5	
Utility 2	0	0.1	0.1	0.1	0.1	
Utility 3	42.6	44	48.4	51.2	54.8	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	10	4.1	13.3	40	32.6	0
U3 - Toxicity prior Beta(0.15, 0.85)	15.2	6	12.6	39.4	26.8	0
U2 - Toxicity prior Beta(0.3, 0.7)	9.9	4.3	12.5	40.5	32.8	0
U3 - Toxicity prior Beta(0.3, 0.7)	14.4	6	11.6	40.2	27.8	0
U2 - Toxicity prior Beta(0.5, 0.5)	10	3.9	11.4	39.2	35.5	0
U3 - Toxicity prior Beta(0.5, 0.5)	13.9	5.4	11.8	39.5	29.4	0
U2 - Toxicity prior Beta(1, 1)	9.3	3.5	11.9	39.3	36	0
U3 - Toxicity prior Beta(1, 1)	13.9	6	11.3	39.2	29.6	0
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	3.4	3.5	5.8	9.7	7.6	
U3 - Toxicity prior Beta(0.15, 0.85)	3.5	3.7	5.8	10.2	6.8	
U2 - Toxicity prior Beta(0.3, 0.7)	3.4	3.5	5.7	9.8	7.7	
U3 - Toxicity prior Beta(0.3, 0.7)	3.5	3.7	5.7	10.1	7.1	
U2 - Toxicity prior Beta(0.5, 0.5)	3.3	3.4	5.5	9.8	8	
U3 - Toxicity prior Beta(0.5, 0.5)	3.4	3.6	5.6	10.1	7.3	
U2 - Toxicity prior Beta(1, 1)	3.3	3.5	5.3	9.6	8.2	
U3 - Toxicity prior Beta(1, 1)	3.4	3.6	5.5	10	7.5	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.1	2.6	1.4	2.7	1.2	
U3 - Toxicity prior Beta(0.15, 0.85)	0.1	2.6	1.5	2.7	1.1	
U2 - Toxicity prior Beta(0.3, 0.7)	0.1	2.6	1.4	2.7	1.1	
U3 - Toxicity prior Beta(0.3, 0.7)	0.1	2.7	1.5	2.7	1.1	
U2 - Toxicity prior Beta(0.5, 0.5)	0.1	2.6	1.4	2.7	1.2	
U3 - Toxicity prior Beta(0.5, 0.5)	0.1	2.6	1.4	2.6	1.1	
U2 - Toxicity prior Beta(1, 1)	0.1	2.6	1.4	2.5	1.1	
U3 - Toxicity prior Beta(1, 1)	0.1	2.6	1.4	2.6	1.1	
Scenario 34						
DLT rate	0.05	0.13	0.3	0.36	0.42	
Efficacy rate	0.05	0.1	0.3	0.32	0.35	
Utility 2	0	0	0	0	-0.1	
Utility 3	41	40.8	46	44.8	44.2	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	16.4	12.7	44.1	20	6.5	0.3
U3 - Toxicity prior Beta(0.15, 0.85)	21.9	14	41.7	18.2	4	0.2
U2 - Toxicity prior Beta(0.3, 0.7)	15.3	12.2	44	20.5	7.8	0.2
U3 - Toxicity prior Beta(0.3, 0.7)	21.3	14	41.3	19.2	4	0.2
U2 - Toxicity prior Beta(0.5, 0.5)	15.1	12	43	21.8	7.9	0.2
U3 - Toxicity prior Beta(0.5, 0.5)	20.9	13.9	41.2	18.4	5.4	0.2
U2 - Toxicity prior Beta(1, 1)	16.4	11.5	40.8	22.1	9	0.2
U3 - Toxicity prior Beta(1, 1)	20.6	13.8	41.1	18.4	5.9	0.2
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	4.9	6	9.1	6.6	3.4	
U3 - Toxicity prior Beta(0.15, 0.85)	5.6	6	9.1	6.3	3	
U2 - Toxicity prior Beta(0.3, 0.7)	4.8	5.7	9.2	6.8	3.5	
U3 - Toxicity prior Beta(0.3, 0.7)	5.3	5.8	9.1	6.6	3.2	
U2 - Toxicity prior Beta(0.5, 0.5)	4.6	5.6	9.1	6.9	3.9	
U3 - Toxicity prior Beta(0.5, 0.5)	5.2	5.7	9	6.6	3.5	
U2 - Toxicity prior Beta(1, 1)	4.5	5.4	8.9	7	4.2	
U3 - Toxicity prior Beta(1, 1)	5	5.5	8.9	6.8	3.8	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.5	3.3	2.1	2	0.8	
U3 - Toxicity prior Beta(0.15, 0.85)	0.6	3.3	2	1.9	0.7	
U2 - Toxicity prior Beta(0.3, 0.7)	0.4	3.3	2	2.1	0.8	
U3 - Toxicity prior Beta(0.3, 0.7)	0.5	3.3	1.9	2	0.7	
U2 - Toxicity prior Beta(0.5, 0.5)	0.4	3.2	1.9	2.1	0.7	
U3 - Toxicity prior Beta(0.5, 0.5)	0.5	3.2	1.9	2	0.7	
U2 - Toxicity prior Beta(1, 1)	0.4	3	1.9	2	0.6	
U3 - Toxicity prior Beta(1, 1)	0.5	3	1.9	1.9	0.6	

Phase I/II Operating Characteristics

Random model - Toxicity prior comparison - allocation = one_stage - IPDE alpha = 0 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 35						
DLT rate	0.03	0.09	0.3	0.4	0.45	
Efficacy rate	0.2	0.45	0.45	0.45	0.45	
Utility 2	0.2	0.4	0.1	0.1	0	
Utility 3	50.8	63.4	55	51	49	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	10.8	37.3	31.3	15.6	5	0
U3 - Toxicity prior Beta(0.15, 0.85)	15.7	43.3	26.2	12	2.8	0
U2 - Toxicity prior Beta(0.3, 0.7)	10.1	37.1	31.2	16	5.6	0
U3 - Toxicity prior Beta(0.3, 0.7)	15.9	43.5	24.9	12.2	3.5	0
U2 - Toxicity prior Beta(0.5, 0.5)	11.4	37.1	30.9	14.5	6.1	0
U3 - Toxicity prior Beta(0.5, 0.5)	15.6	40.9	27.5	12.1	3.9	0
U2 - Toxicity prior Beta(1, 1)	10.4	36.1	30.5	16.9	6.1	0
U3 - Toxicity prior Beta(1, 1)	14.5	41.1	26.8	13.5	4.1	0
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	4.4	9.9	8.5	5.1	2.2	
U3 - Toxicity prior Beta(0.15, 0.85)	4.6	11.1	8.1	4.4	1.6	
U2 - Toxicity prior Beta(0.3, 0.7)	4.3	10	8.4	5	2.3	
U3 - Toxicity prior Beta(0.3, 0.7)	4.5	11	8.1	4.6	1.8	
U2 - Toxicity prior Beta(0.5, 0.5)	4.2	9.8	8.3	5.3	2.4	
U3 - Toxicity prior Beta(0.5, 0.5)	4.4	10.7	8.2	4.8	1.9	
U2 - Toxicity prior Beta(1, 1)	4.2	9.7	8.1	5.5	2.5	
U3 - Toxicity prior Beta(1, 1)	4.4	10.8	8.1	4.8	1.9	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.2	2.9	1.8	1.4	0.4	
U3 - Toxicity prior Beta(0.15, 0.85)	0.2	3	1.8	1.2	0.3	
U2 - Toxicity prior Beta(0.3, 0.7)	0.2	2.9	1.8	1.3	0.4	
U3 - Toxicity prior Beta(0.3, 0.7)	0.2	3	1.7	1.2	0.3	
U2 - Toxicity prior Beta(0.5, 0.5)	0.1	2.9	1.8	1.3	0.5	
U3 - Toxicity prior Beta(0.5, 0.5)	0.2	2.9	1.8	1.2	0.4	
U2 - Toxicity prior Beta(1, 1)	0.1	2.8	1.7	1.3	0.4	
U3 - Toxicity prior Beta(1, 1)	0.2	2.9	1.7	1.2	0.3	
Scenario 36						
DLT rate	0.13	0.29	0.35	0.4	0.43	
Efficacy rate	0.05	0.2	0.22	0.3	0.35	
Utility 2	-0.1	-0.1	-0.1	-0.1	-0.1	
Utility 3	37.8	40.4	39.2	42	43.8	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	22.1	44.1	16.7	9.3	3.4	4.4
U3 - Toxicity prior Beta(0.15, 0.85)	26.5	43.9	14.6	8	1.8	5.2
U2 - Toxicity prior Beta(0.3, 0.7)	20.1	45.3	17.3	9.6	3.7	4
U3 - Toxicity prior Beta(0.3, 0.7)	25.8	45.4	14.2	8.5	1.8	4.3
U2 - Toxicity prior Beta(0.5, 0.5)	18.3	44.4	18.4	10.9	4.4	3.6
U3 - Toxicity prior Beta(0.5, 0.5)	25.3	43.4	15.8	8.7	2.8	4
U2 - Toxicity prior Beta(1, 1)	17.6	41.9	18.4	12.4	5.8	3.9
U3 - Toxicity prior Beta(1, 1)	25	42.2	15.3	10.4	2.6	4.5
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	8	9.7	5.9	4.1	1.8	
U3 - Toxicity prior Beta(0.15, 0.85)	8.6	9.6	5.7	3.9	1.7	
U2 - Toxicity prior Beta(0.3, 0.7)	7.5	9.5	6.2	4.4	2.1	
U3 - Toxicity prior Beta(0.3, 0.7)	8.1	9.5	5.9	4.2	1.8	
U2 - Toxicity prior Beta(0.5, 0.5)	7.1	9.4	6.4	4.6	2.3	
U3 - Toxicity prior Beta(0.5, 0.5)	7.7	9.4	6.1	4.4	2.1	
U2 - Toxicity prior Beta(1, 1)	7	9.2	6.3	4.7	2.4	
U3 - Toxicity prior Beta(1, 1)	7.7	9	6.1	4.6	2.2	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.8	4.3	1.6	1.3	0.4	
U3 - Toxicity prior Beta(0.15, 0.85)	0.9	4.3	1.5	1.2	0.3	
U2 - Toxicity prior Beta(0.3, 0.7)	0.8	4.3	1.6	1.3	0.4	
U3 - Toxicity prior Beta(0.3, 0.7)	0.9	4.1	1.5	1.2	0.3	
U2 - Toxicity prior Beta(0.5, 0.5)	0.7	3.9	1.4	1.3	0.4	
U3 - Toxicity prior Beta(0.5, 0.5)	0.8	3.9	1.4	1.2	0.3	
U2 - Toxicity prior Beta(1, 1)	0.6	3.5	1.4	1.2	0.3	
U3 - Toxicity prior Beta(1, 1)	0.7	3.5	1.4	1.1	0.3	

Phase I/II Operating Characteristics

Random model - Toxicity prior comparison - allocation = one_stage - IPDE alpha = 0 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 37						
DLT rate	0.08	0.3	0.4	0.45	0.5	
Efficacy rate	0.05	0.2	0.2	0.2	0.2	
Utility 2	0	-0.1	-0.2	-0.2	-0.3	
Utility 3	39.8	40	36	34	32	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	22.9	51.9	16.8	4.8	1.1	2.5
U3 - Toxicity prior Beta(0.15, 0.85)	29.1	52.2	12.1	3.5	0.5	2.6
U2 - Toxicity prior Beta(0.3, 0.7)	23	50.9	16.8	5.5	1.4	2.4
U3 - Toxicity prior Beta(0.3, 0.7)	28.7	50.6	14.4	3.8	0.6	1.9
U2 - Toxicity prior Beta(0.5, 0.5)	21.4	51.3	17.9	5.7	2	1.7
U3 - Toxicity prior Beta(0.5, 0.5)	29.5	49.1	14.6	3.9	0.8	2.1
U2 - Toxicity prior Beta(1, 1)	22.5	49.7	17.8	5.8	1.9	2.3
U3 - Toxicity prior Beta(1, 1)	27.4	50.3	14.9	4.2	0.7	2.5
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	7.8	10.9	6.2	3.6	1.4	
U3 - Toxicity prior Beta(0.15, 0.85)	8.5	10.8	5.9	3.5	1.2	
U2 - Toxicity prior Beta(0.3, 0.7)	7.5	10.6	6.3	3.9	1.5	
U3 - Toxicity prior Beta(0.3, 0.7)	8.2	10.5	6.1	3.7	1.4	
U2 - Toxicity prior Beta(0.5, 0.5)	7.2	10.5	6.5	4	1.7	
U3 - Toxicity prior Beta(0.5, 0.5)	7.8	10.4	6.2	3.9	1.5	
U2 - Toxicity prior Beta(1, 1)	7	10.3	6.5	4.2	1.8	
U3 - Toxicity prior Beta(1, 1)	7.7	10.3	6.2	4.1	1.6	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.9	4.7	1.7	1.2	0.3	
U3 - Toxicity prior Beta(0.15, 0.85)	1	4.6	1.7	1.2	0.3	
U2 - Toxicity prior Beta(0.3, 0.7)	0.9	4.6	1.6	1.2	0.3	
U3 - Toxicity prior Beta(0.3, 0.7)	1	4.4	1.6	1.1	0.3	
U2 - Toxicity prior Beta(0.5, 0.5)	0.9	4.3	1.5	1.2	0.3	
U3 - Toxicity prior Beta(0.5, 0.5)	0.9	4.2	1.5	1.1	0.3	
U2 - Toxicity prior Beta(1, 1)	0.8	3.9	1.4	1.1	0.3	
U3 - Toxicity prior Beta(1, 1)	1	3.9	1.4	1.1	0.2	